A LAUNDRY TRACKING SYSTEM

SYSTEM DOCUMENTATION

SUBMITTED BY,

AMINA ABDIRIZAK ARALE

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DECLARATION

The content of this document is product of my own personal research as guided by unit lecturer and following the provided standards of system documentation.

Student Name: _____ Sign: _____ Date: _____

Supervisor Name: _____ Sign: _____ Date: _____

DEDICATION

Firstly, I would like to thank my unit lecturer, Mr. Victor Otieno for offering the much support and guidance I needed during the system creation period. Secondly, I would also like to thank my project supervisor Mr. Barrack Ouko for being there for me, his support and generosity during the project period. Lastly, I would like to give my appreciation to my other college students for their support during this long journey and their assists during my project`s creation time

ABSTRACT

The Laundry Business Tracker System is a web-based application designed to streamline and enhance the management of daily operations in small to medium-sized laundry businesses. The system provides an efficient, reliable, and user-friendly platform for recording, tracking, and analyzing business activities, enabling business owners to maintain accurate records and make informed decisions.

The application allows users to register new customer orders by capturing key details such as the type of service provided, payment method, and payment status. It supports multiple payment options, including M-Pesa and cash, and enables tracking of whether payments are completed or pending. This functionality improves financial accountability and minimizes errors associated with manual record-keeping.

Additionally, the system incorporates an expense management module that allows users to record and monitor business expenses. This feature provides a clear overview of operational costs, supporting effective budgeting and financial planning. A dashboard is also included to visualize financial performance through charts and summaries, allowing users to quickly interpret business trends and overall performance.

The system is developed using HTML, CSS, and JavaScript for the frontend, ensuring a responsive and interactive user interface. Supabase is utilized for backend services, including authentication and database management, providing secure data storage and real-time data

DEFINITION OF KEY TERMS

Laundry Business Tracker System: A web-based application designed to manage and monitor daily laundry operations such as order registration, payment tracking, and expense management.

Order Registration: The process of recording a new laundry service request, including customer details, type of service, and payment information.

Payment Tracking: The monitoring of transactions to determine whether a customer has paid for services, including tracking payment methods such as M-Pesa or cash.

M-Pesa Payment: A mobile money payment method that allows customers to pay for laundry services electronically through their mobile devices.

Cash Payment: A traditional payment method where customers pay physically using cash for laundry services rendered.

Paid Status: A system indicator showing that a customer has fully settled the payment for a laundry service.

Unpaid Status: A system indicator showing that a customer has not yet completed payment for a laundry service.

Expense Management: The process of recording, tracking, and analyzing business expenses such as detergents, water, electricity, and labor costs.

Dashboard: A visual interface that displays key business data such as total sales, expenses, and payment status using charts and summaries.

Data Visualization: The graphical representation of business data through charts and graphs to help users easily understand financial performance.

User Authentication: A security process managed by the system to verify the identity of users before granting access to the application.

Database: A structured collection of data used to store information such as orders, payments, and expenses in the system.

Supabase: A backend platform used in the system to provide database services, authentication, and real-time data management.

Frontend: The user interface of the system developed using HTML, CSS, and JavaScript, which users interact with directly.

Backend: The server-side component of the system responsible for data processing, storage, and authentication, implemented using Supabase.

Web Application: A software application that runs on a web browser, allowing users to access the system online without installing additional software.

Real-Time Data: Information that is updated instantly within the system, ensuring users always view the most current records.

User Interface (UI): The visual layout and design of the system that enables user interaction with features such as forms, buttons, and dashboards.

System Implementation: The process of deploying the laundry tracking system for actual use, including setup, configuration, and user onboarding.

Efficiency: The ability of the system to perform tasks quickly and accurately, reducing manual work and improving productivity.

Convenience: The ease of use provided by the system, allowing users to manage laundry operations seamlessly from a web browser.

Webpage: A single document or page within the system that displays specific information, such as the dashboard, orders page, or expenses page.

ABBREVIATIONS AND ACRONYMS

HTML – Hyper Text Markup Language

CSS – Cascading Style Sheets

JS – JavaScript

UI – User Interface

DB – Database

API – Application Programming Interface

MPESA – Mobile Payment System (M-Pesa)

CRUD – Create, Read, Update, Delete

URL – Uniform Resource Locator

HTTP – Hyper Text Transfer Protocol

HTTPS – Hyper Text Transfer Protocol Secure

JSON – JavaScript Object Notation

SQL – Structured Query Language

BaaS – Backend as a Service

SaaS – Software as a Service

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CHAPTER ONE: PROJECT PLANNING AND ANALYSIS

1.1 Statement of Problem

Many small laundry businesses face challenges in managing their daily operations efficiently. Most of these businesses rely on manual record-keeping methods such as notebooks or memory to track customer orders, payments, and expenses. This often leads to errors such as lost records, incorrect payment tracking, and poor financial management.

In addition, it becomes difficult for the business owner to clearly determine which orders have been paid for and which are still pending. Tracking payments made through different methods such as M-Pesa and cash can also be confusing without a proper system in place. Furthermore, managing expenses manually makes it hard to monitor business profitability and control costs effectively.

The lack of a centralized system also means that business performance cannot be easily analyzed, as there are no visual summaries or reports. This results in poor decision-making and reduced efficiency in the overall management of the laundry business.

1.2 Study Justification

The Laundry Business Tracker System provides an effective solution to the challenges faced in manual laundry business management. The system improves efficiency by allowing the admin to digitally record and manage customer orders, payments, and expenses in one centralized platform.

By incorporating payment tracking features, the system ensures that all transactions—whether through M-Pesa or cash—are accurately recorded and monitored. This reduces errors and improves financial accountability. The expense management feature also enables the admin to track operational costs, making it easier to manage budgets and increase profitability.

Additionally, the system includes a dashboard that provides visual insights into business performance through charts and summaries. This helps the admin quickly understand trends, monitor income and expenses, and make informed decisions.

Overall, the system enhances convenience, accuracy, and efficiency, making it a valuable tool for modernizing laundry business operations.

1.3 System Objectives

1.3.1 General Objective

The main aim of this project is to develop a Laundry Business Tracker System that helps in managing orders, payments, and expenses efficiently.

1.3.2 Specific Objectives

(i) To develop an interface that enables the admin to register and manage laundry orders.

(ii) To create a system that allows tracking of payments (M-Pesa or cash) and their status (paid or unpaid).

(iii) To develop an expense management module for recording and monitoring business expenses.

(iv) To create a dashboard that displays business performance using charts and summaries.

1.4 Functional Requirements

User	User Activities	Features
Admin	- Login - Access dashboard - Register new laundry orders - Record payment method (M-Pesa or cash) - Update payment status (paid/unpaid) - Manage and track expenses - View reports and summaries - Logout	- Secure login and logout - Order management system - Payment tracking (M-Pesa & cash) - Expense management - Dashboard with charts and summaries - Real-time data updates - Data storage and retrieval (Supabase) - Simple and user-friendly interface

Table 1.4 Functional Requirements Table

1.5 Breakdown of Tools & Resources to Be Used

The system will be developed using Python with the Flask framework for the backend, and HTML, CSS, and JavaScript for the frontend. SQLite will be used as the database to store user information and detection history. Additional tools include libraries for handling images, sending emails, and processing requests. The system will be developed using VS Code and deployed on an online hosting platform.

1.6 Project Schedule Breakdown

	PROJECT MILESTONES					
WEEKS	Project Planning & Analysis (System Documentation: Cover page & Chapter One)	Project Design & Modeling (System Documentation Chapter Two)	Project Development & Testing (System Documentation Chapter Three)	Project Deployment (System Documentation Chapter Three)	Final Touches of System Documentation (Preliminary Pages, Chapter Four & References)	Project Presentation
5-9 May						
12-16 May						
19-23 May						
26-30 May						
2-6 June						
9-13 June						
16-20 June						
23-27 June						
30-4 July						
7-11 July						
14-18 July						
21-25 July						
11 Aug						

Table 1.6 Project Schedule Breakdown

CHAPTER TWO: DESIGN AND MODELING

2.1 Introduction to Modelling

This chapter presents the design and modeling of the Laundry Business Tracker System. The system is designed exclusively for the laundry owner to manage all operations efficiently. The owner can create and track orders, monitor paid and unpaid orders, manage expenses, view analytics, and handle customer reviews. The system provides a centralized dashboard to oversee daily business activities.

2.2 User Interface Models

2.2.1 **New Orders Page**

The New Orders Page allows the owner to record new laundry orders. Each order includes:

- **Order ID / Customer Name**
- **Description of the Cloth**
- **Number of Pieces**
- **Price per Piece**
- **Total Amount**

This page enables accurate tracking of incoming laundry requests.

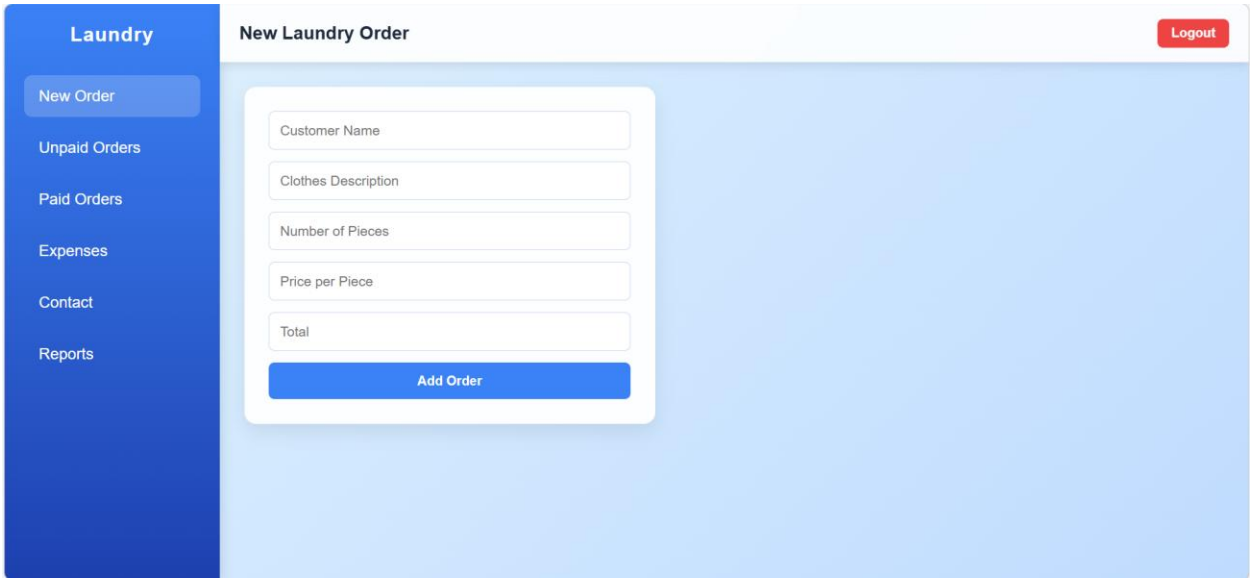

Figure 2.2.1: New orders page

2.2.2 Unpaid orders page

The Unpaid Orders Page lists all pending payments. Each entry includes:

- **Order Number / Customer Name**
- **Description of the Cloth**
- **Number of Pieces**
- **Total Amount**
- **Action** (Payment method: Mpesa or Cash)

This page helps the owner monitor which orders are yet to be paid and manage follow-ups.

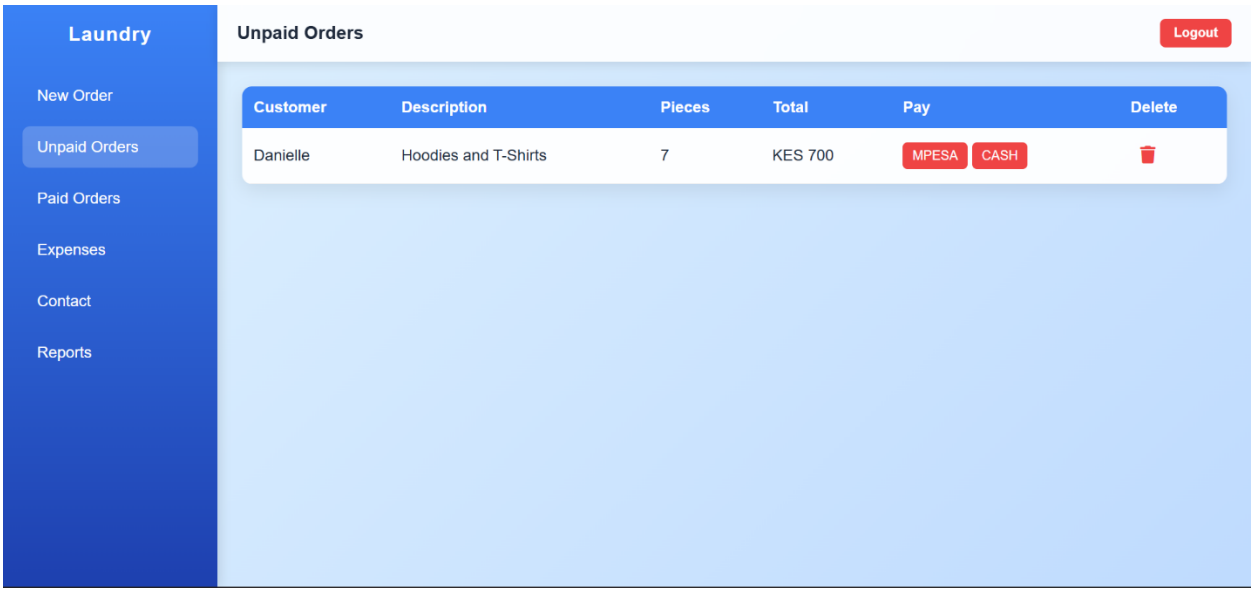

Figure 2.2.2: Unpaid orders page

2.2.3 Paid orders page

The Paid Orders Page displays all completed and paid orders. Each entry includes:

- **Customer Name**
- **Description of the Cloth**
- **Payment Method**
- **Total Amount**

This page allows the owner to keep a record of completed transactions.

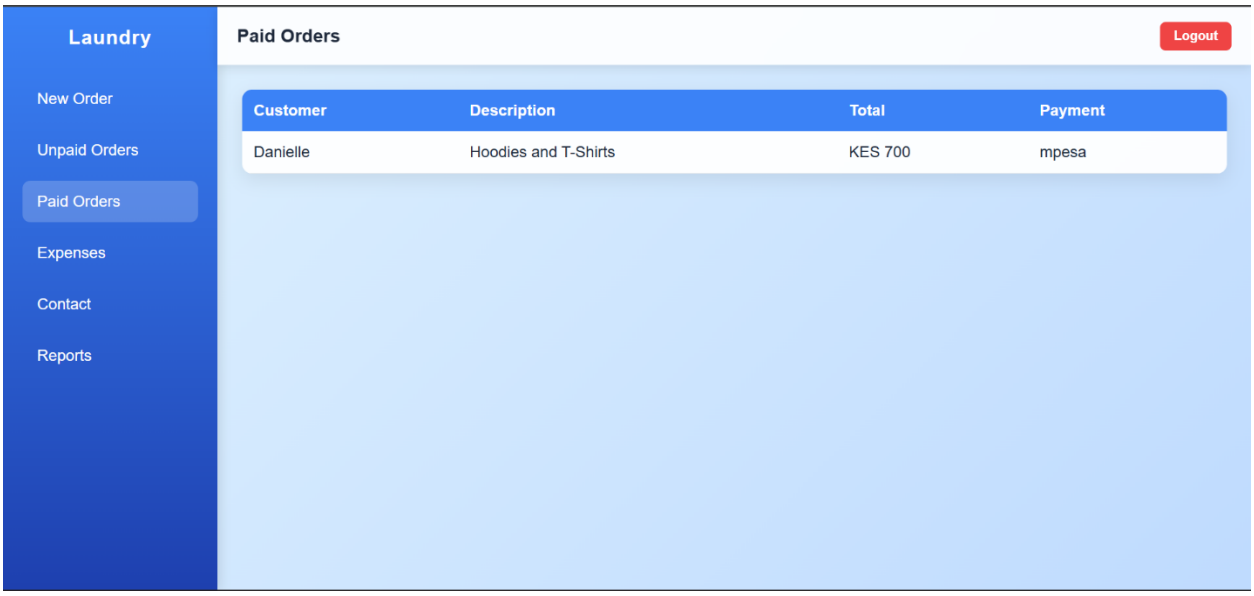

Figure 2.2.3: Paid orders page

2.2.4 Contact / Review page

The Contact Page allows the owner to view customer reviews and maintain contact information. Features include:

- **Customer Name**
- **Rating (1–5)**
- **Review Comments**
- **Contact Information**

This page helps the owner gather feedback to improve service quality.

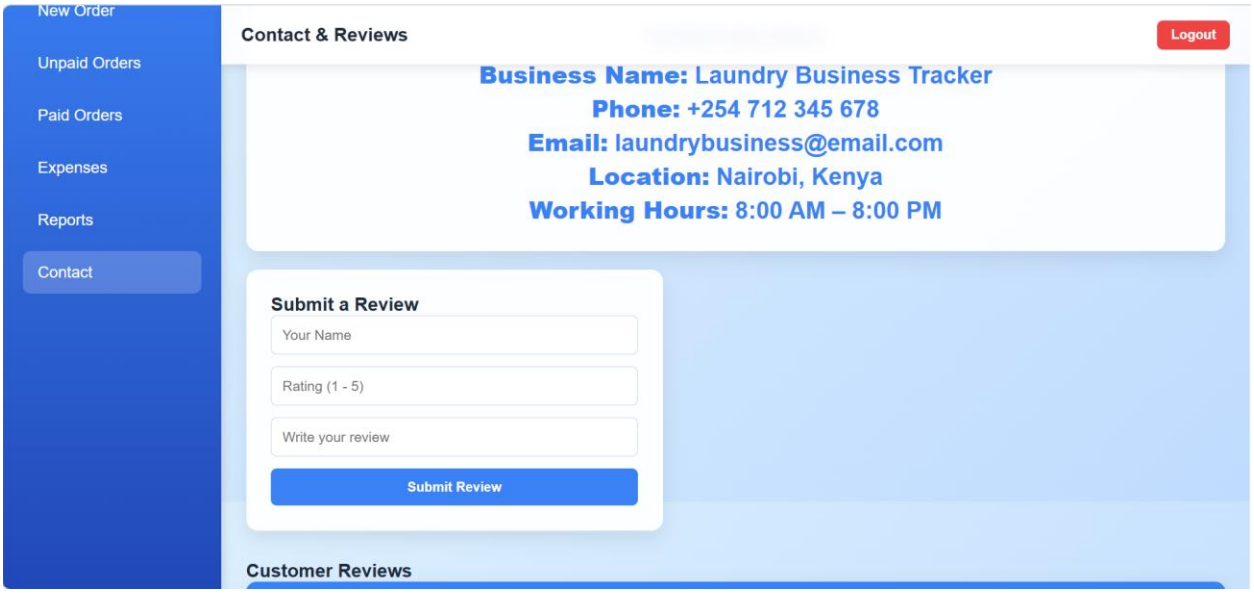

Figure 2.2.4: Contact / Review page

2.2.5 **Expense and Analytics Page**

The Expense and Analytics Page provides a financial overview of the laundry business. It includes:

- **Revenue**
- **Profit**
- **Expenses**
- **A section to add new expenses with their amounts**

This page enables the owner to make informed financial decisions and track the overall profitability of the business.

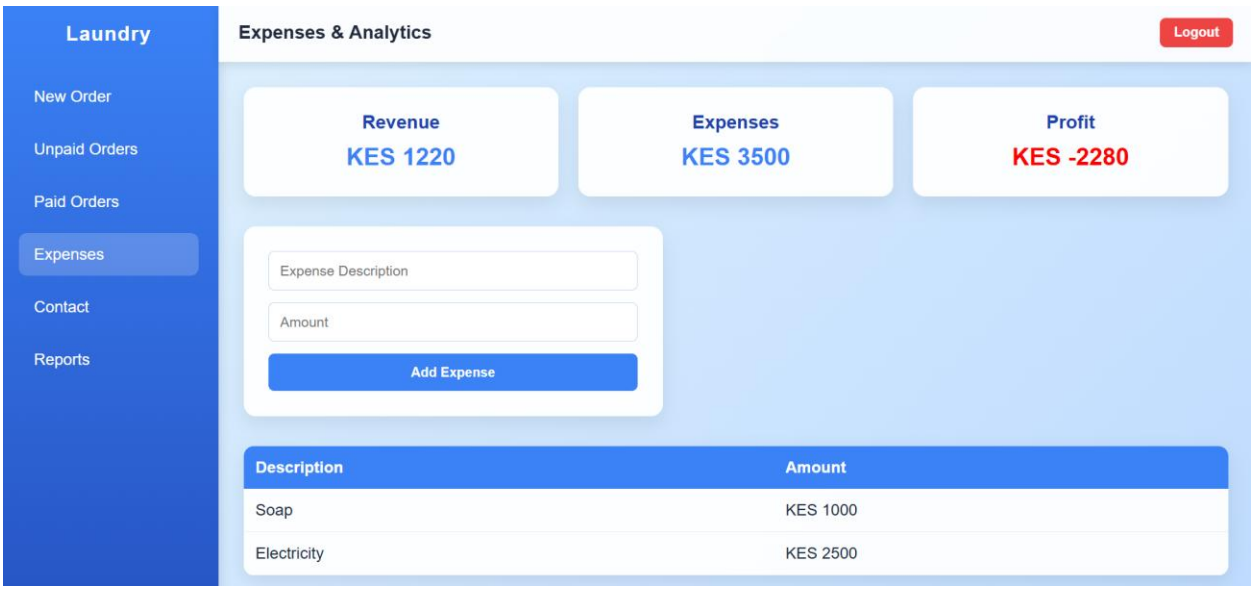

Figure 2.2.5: Expense and analytic page

2.3 Logic Models

2.3.1 Order Processing Flowchart

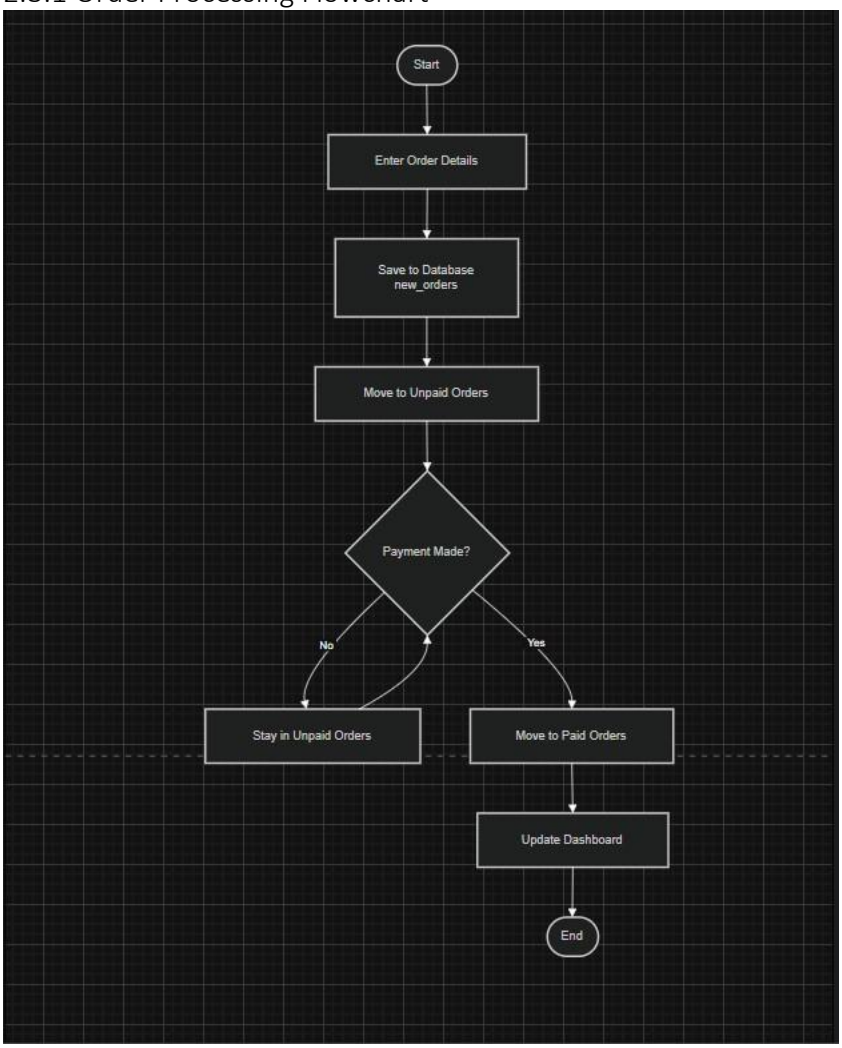

Figure 2.3.1: Order Processing Flowchart

2.3.2 **Payment Processing Flowchart**

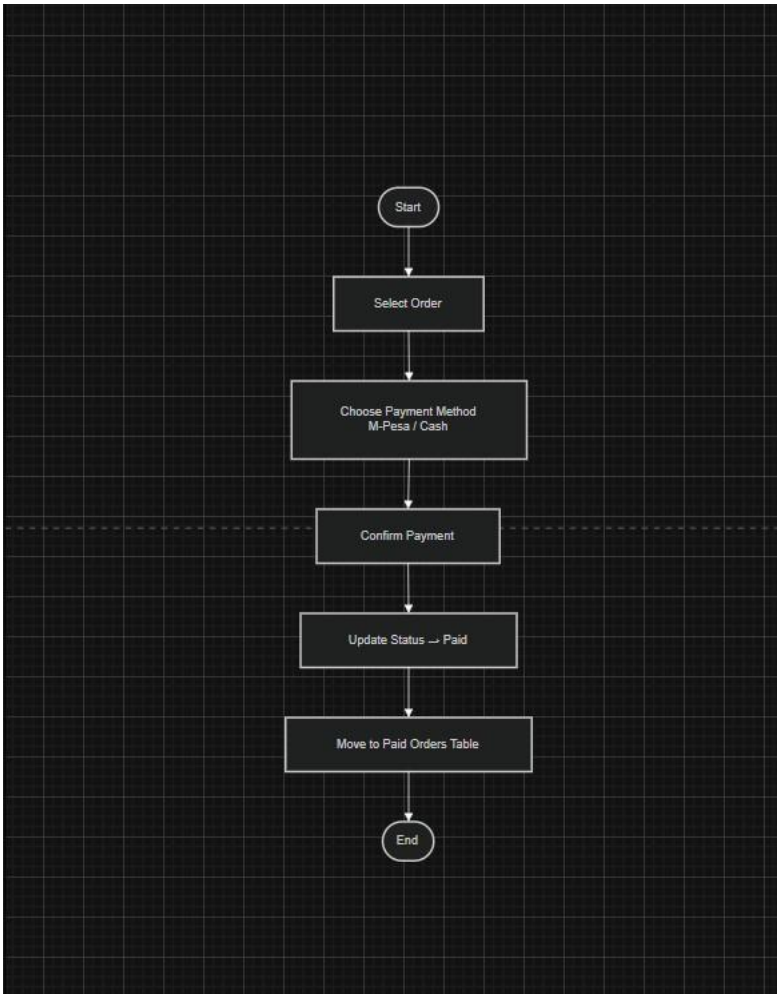

Figure 2.3.2: **Payment Processing Flowchart**

2.3.3 System Flow Diagram

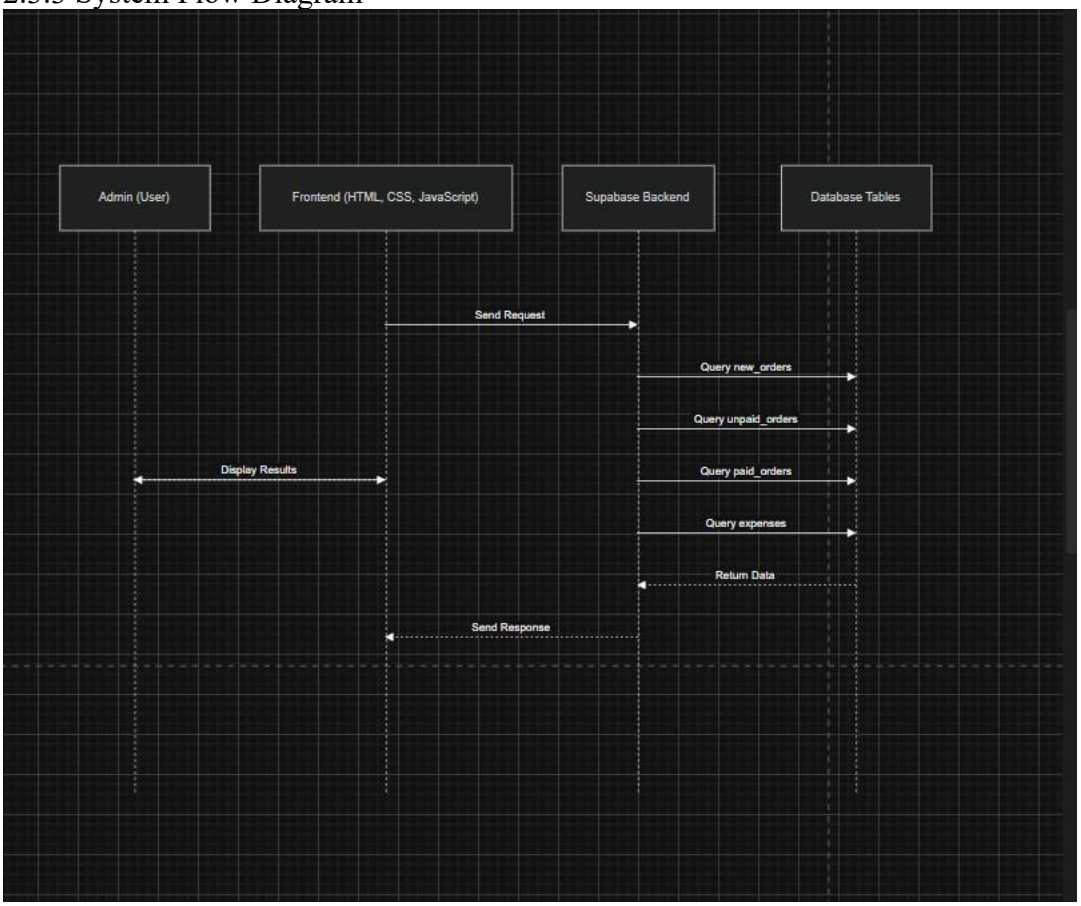

Figure 2.3.3: System Flow Diagram

2.3.4 Database Structure Diagram

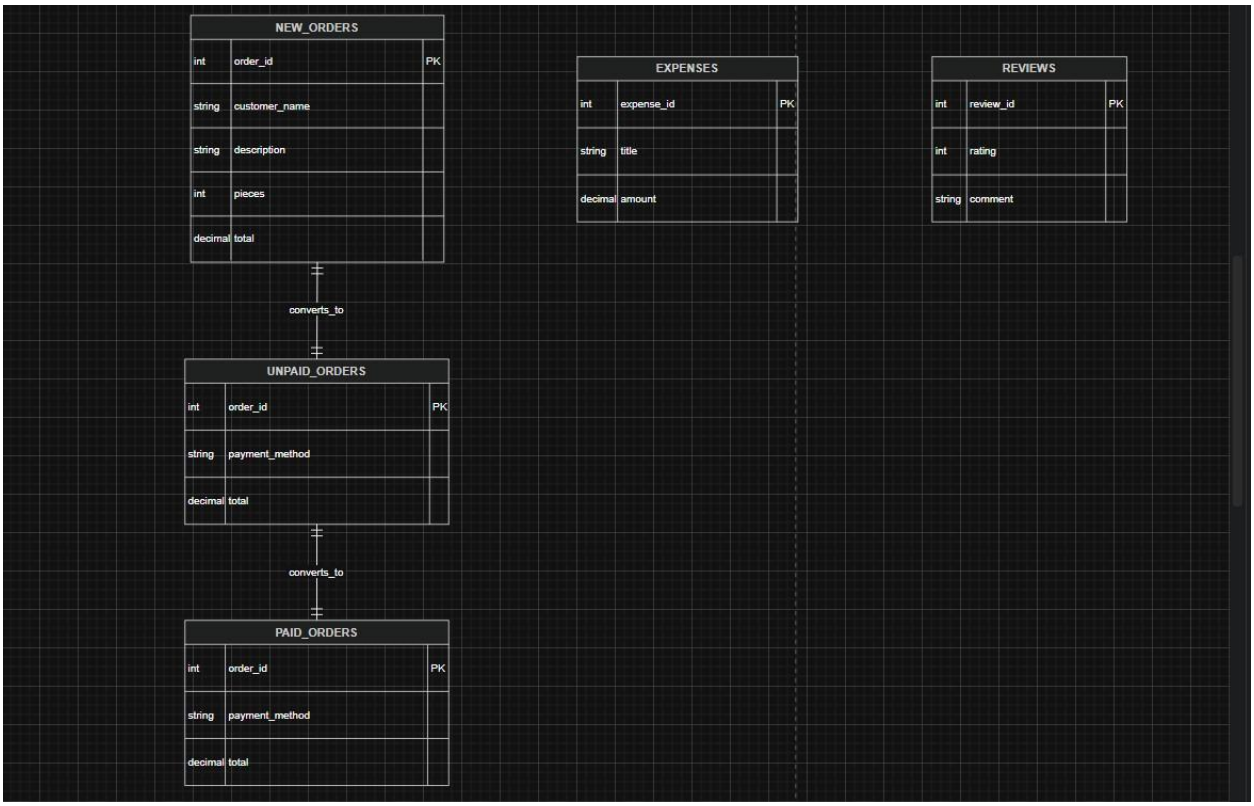

Figure 2.3.4: Database Structure Diagram

CHAPTER THREE: SYSTEM IMPLEMENTATION

3.1 Introduction

This chapter focuses on the development, testing, and deployment of the **Laundry Business Tracker System**. The system is designed for the laundry owner to manage orders, track paid and unpaid orders, monitor expenses, analyze revenue and profit, and view customer reviews. It includes the code logic connecting the user interface to the Supabase backend, tests conducted to ensure system performance, and the deployment setup.

3.2 User Interface Development

This section shows the development of the user-facing part of the system, including the pages users interact with such as login, signup, dashboard, and analysis pages.

3.3 Logic Development

This section shows how the system processes data and performs operations such as login validation, media analysis, and database storage.

3.4 Testing

The Laundry Business Tracker System was tested for functionality, usability, and performance:

Testing Scenarios Conducted:

- Adding new orders and updating unpaid orders.
- Marking orders as paid and verifying paid orders list.
- Adding expenses and verifying profit/revenue calculations.
- Submitting and viewing customer reviews.
- Secure login for the laundry owner.
- Responsive UI on multiple browsers (Chrome, Firefox, Edge).

Test Results Summary:

- All core functionalities worked as intended.
- Orders, payments, reviews, and expenses stored correctly in Supabase.
- Dashboard displays accurate summaries.
- No critical bugs were found.
- System is stable under normal usage conditions.

Test Case	Expected Result	Actual Result	Status
Login	Access granted	Works correctly	Pass
Add Order	Saved in database	Works correctly	Pass
Payment Update	Status updated	Works correctly	Pass
Add Expense	Recorded	Works correctly	Pass
Dashboard	Displays correct data	Works correctly	Pass

3.5 Deployment

The Laundry Business Tracker System was deployed as a web-based application using modern web technologies. The frontend was developed using HTML, CSS, and JavaScript, while Supabase was used as the backend service for authentication and database management.

System URL: https://laundry-business-tracker.vercel.app/

CHAPTER FOUR: CONCLUSION AND RECOMMENDATION

4.1 Conclusion

In conclusion, the Laundry Business Tracker System has effectively achieved its goal of improving the management and tracking of laundry operations. By providing a centralized platform for monitoring new orders, unpaid and paid orders, customer reviews, expenses, and analytics, the system enhances efficiency, reduces errors, and supports better decision-making for the laundry owner. Its user-friendly interface allows the owner to manage all aspects of the business from a single dashboard, ensuring timely processing of orders and accurate financial tracking. The automation of order management, payment tracking, and expense calculation reduces reliance on manual record-keeping, providing greater accuracy, consistency, and reliability in daily operations. Moving forward, the system can be further enhanced by incorporating features such as automated reminders for unpaid orders, mobile notifications for customers, and integration with additional payment gateways to provide a more comprehensive and adaptive solution for modern laundry business needs.

4.2 Recommendation

To maximize the effectiveness and impact of the Laundry Business Tracker System, it is recommended to implement the following strategies:

1. Enhance Customer Feedback: Integrate additional customer feedback channels, such as in-app surveys or comment forms, to collect insights on service quality. This will allow the owner to continuously improve service delivery.

2. Awareness and Training: Conduct training sessions or provide guides for staff (if any) to ensure proper usage of the system. Awareness campaigns for customers, highlighting features such as reviews submission and service tracking, can also encourage participation.

3. Payment Integration: Expand payment options to include more mobile money platforms, bank transfers, or card payments, making the system more convenient for customers and streamlining revenue collection.

4. Analytics Enhancement: Introduce advanced reporting features such as trend analysis for revenue, seasonal order patterns, and expense optimization. This will help the owner make data-driven decisions to improve profitability.

5. System Maintenance: Schedule regular maintenance and updates to ensure the system remains secure, responsive, and compatible with evolving web technologies.

6. Scalability: Consider adding features for future growth, such as multiple branch management, staff performance tracking, and automated notifications for customers and staff, to expand the system's capabilities as the business grows.

By adopting these strategies, the Laundry Business Tracker System can continue to provide a reliable, user-friendly, and forward-looking solution for efficient laundry business management.

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